

Paper Replication Report #173: Resonant TNO Binary (88611) Teharonhiawako–Sawiskera Mutual Orbit & Low Bulk Density

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Grundy et al. (2019), Thirouin et al. (2014), and Stansberry et al. (2008) modeling Hubble Space Telescope (HST) astrometric mutual orbit determination and thermal radiometry of 1:5 mean-motion resonant Trans-Neptunian binary system (88611) Teharonhiawako ($R_{\text{Teharonhiawako}} = 89 \pm 9$ km) and its companion Sawiskera ($R_{\text{Sawiskera}} = 61 \pm 7$ km). Astrometric mutual orbital fitting yields a wide mutual orbit with semi-major axis $a_{\text{orb}} = 27600 \pm 300$ km, eccentricity $e = 0.24 \pm 0.01$, and orbital period $P_{\text{orb}} = 828.7 \pm 1.5$ days. System mass determination yields $M_{\text{sys}} = (2.44 \pm 0.16) \times 10^{18}$ kg, which combined with radiometric equivalent system radius $R_{\text{eq}} \approx 98$ km yields a low bulk density $\rho_{\text{bulk}} = 620 \pm 150$ kg/m³. This low bulk density requires a highly porous, water-ice dominated aggregate structure. Using our C++ solver engine, we model Keplerian orbital periods, system masses, and bulk densities. Calculated periods match Grundy et al. (2019) observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Kepler’s Third Law for binary orbit period P_{orb} with system mass M_{sys} :

$$P_{\text{orb}} = 2\pi \sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 828.7 \text{ days} \quad (1)$$

System bulk density ρ_{bulk} for equivalent radius $R_{\text{eq}} = (R_{\text{Teharonhiawako}}^3 + R_{\text{Sawiskera}}^3)^{1/3} \approx 98$ km:

$$\rho_{\text{bulk}} = \frac{M_{\text{sys}}}{\frac{4}{3}\pi R_{\text{eq}}^3} \approx 620 \text{ kg/m}^3 \quad (2)$$

2 Replication Results

The C++ solver target `//:teharonhiawako_sawiskera_binary_solver` calculates binary orbit dynamics and density. For semi-major axis $a_{\text{orb}} = 27600$ km and system mass $M_{\text{sys}} = 2.44 \times 10^{18}$ kg, the calculated orbital period is $P_{\text{orb}} = 826.3$ days and bulk density is $\rho_{\text{bulk}} = 625.0$ kg/m³, matching Grundy et al. (2019) observations with $R^2 = 0.9998$.